

JEE MAIN REVISION MASTER: BIOMOLECULES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 198 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Biomolecules - Core Application	198 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Biomolecules - Core Application (Very High Yield — 198 PYQs)

Core Principle: Key problem pattern in Biomolecules.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The incorrect statement from the following with respect to carbohydrates is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : and are two anomers of D-(+)-glucose. Statement II : The open chain forms o..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the List-I with List-II List-I Name of amino acid List-II One letter symbol/type A. Arginine I. D/Non-essential B. Aspartic acid II. R..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Match the List-I with List-II List-I Name of amino acid List-II One letter symbol/type A. Arginine I. D/Non-essential B. Aspartic acid II. R/Essential C. Lysine III. E/Non-essential D. Glutamic acid IV. K/Essential Choose the correct answer from the options given below :

- (A) -II, B-I, C-IV, D-III **[Correct]**
- (B) -IV, B-III, C-II, D-I
- (C) -III, B-IV, C-I, D-II
- (D) -II, B-IV, C-I, D-III

Correct Answer: A

Solution Analysis: Match each amino acid with its correct one-letter symbol and whether it is essential or non-essential: Arginine → one-letter symbol is R In standard school-level classification here, it is matched as Essential So, A → II Aspartic acid → one-letter symbol is D It is Non-essential So, B → I Lysine → one-letter symbol is K It is Essential So, C → IV Glutamic acid → one-letter symbol is E It is Non-essential So, D → III Thus the correct matching is: A-II,B-I,C-IV,D-III So, the correct option is